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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

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Fifth Semester

Discipline Specific Core Course (DSC) -MG5DSCIAA300

MATERIAL SPECIFICATION -II

PART -A

1. Epoxy resin is a versatile thermosetting polymer that forms a rigid, high-strength material when mixed with a curing agent.
2. Weaving is the craft or process of making fabric and objects by interlacing two distinct sets of yarns or threads at right angles, typically using a device called a loom.
3. Comparing cork and animal leather reveals distinct differences across renewable resources, carbon footprints, manufacturing chemicals, and end-of-life breakdown
4. A thermosetting plastic (or thermoset) is a polymer that irreversibly hardens and sets into a permanent shape when heated and cured.
5. Industrial wiping rags and mechanical shop towels are the most scalable and practical commercial reuse applications for discarded, low-grade textiles. This application diverts high volumes of fabric from landfills without requiring complex fibre separation or chemical recycling.
6. Electrical conductivity is a measure of a material's intrinsic ability to allow an electric current to flow through it.
7. Adhesives are non-metallic substances used to bind two or more separate surfaces together and resist their separation.
8. Five common flooring materials include hardwood, vinyl, laminate, tile, and carpet.
9. Leather requires regular, specialized maintenance to prevent drying and cracking, whereas Rexine (artificial leather/vinyl) is highly low-maintenance and only needs basic cleaning.
10. Mosaic art uses a variety of colourful and textured materials, ranging from manufactured glass to natural stone and recycled household items.
11. Natural flooring is made directly from raw, unedited materials like solid wood, stone, marble, granite, cork, or plant fibres.
12. Skylights deliver up to three times more light than vertical windows of the same size. They bring consistent, ambient daylight deeper into the home's footprint, eliminating dark corners and reducing the need for artificial lighting during the day.

PART -B

13. For a high-end interior project, a cohesive metalwork (architectural metalwork) solution should balance slim-profile acoustics, thermal performance, and continuous structural durability. Moving away from traditional wood towards precision-engineered steel, stainless steel (SS), or architectural aluminium profiles provides an elegant, minimalist look while maximizing glass surface areas. For modern, durable, and aesthetically pleasing interior metalwork, a hybrid Aluminium and Steel solution offers the best balance of structural integrity, weight distribution, and visual appeal. Surface Preparation and Cutting, Assembly and Joining (Welding & Fastening), Anti-Corrosion & Surface FinishingExecution & Installation Methodology

14. Adhesives provide uniform stress distribution, weight reduction, and the ability to join dissimilar materials without damaging the substrates Even Stress Distribution: Unlike mechanical fasteners (like screws, rivets, or bolts) that concentrate stress at specific points, adhesives spread loads evenly across the entire bonded area, increasing structural integrity and fatigue resistance Bonding Dissimilar Materials: They effectively join different or hard-to-bond materials (such as metal to plastic, glass to wood, or carbon fibre to aluminium) without causing bimetallic or galvanic corrosion. Weight Reduction & Thin Substrates: Adhesives eliminate the need for heavy brackets, overlapping joints, or drilled holes, helping reduce weight in automotive, aerospace, and manufacturing applications Preservation of Material Properties: Unlike welding or soldering, adhesive bonding requires minimal or no high heat input, preventing thermal distortion, warping, or changes to the metallurgical structure of the parts.
15. Metal Sheets-Steel sheets coated with a layer of zinc to prevent rust; rigid, durable, and handles heavy rain or wind. Lifespan is roughly 8–15 years. Budget-friendly roofs, rural housing, agricultural sheds, and basic warehouses.
    Colour-Coated (PPGI / PPGL) Sheets, Aluminium Sheets, Galvanized Iron (GI) Sheets, Galvalume Sheet, Plastic and Polymer Sheets, Polycarbonate Sheets, Fiber-Reinforced Plastic (FRP / GRP) Sheet, PVC / uPVC Sheet, Cement-Based Sheet, Fibre Cement (Non-Asbestos) Sheets, Bituminous and Asphalt Sheets Insulated Sandwich Sheets

16. Superior Temperature Resistance: Synthetic rubbers (such as EPDM and silicone) can withstand extreme heat and severe cold without melting or becoming brittle. Chemical and Oil Resistance: Many types—like Acrylonitrile Butadiene Rubber (NBR)—resist oils, fuels, solvents, acids, and alkalis, making them ideal for industrial and automotive fluids. Weather and Ozone Durability: Synthetic rubber resists degradation from sunlight, ultraviolet (UV) rays, oxygen, and ozone, making it long-lasting in outdoor environments (such as roofing materials and gaskets). High Abrasion and Wear Resistance: It often provides greater toughness, strength, and resistance to physical wear, tearing, and friction in heavy-stress applications like vehicle tires. Consistent Quality and Supply: Because it is manufactured in controlled factory settings rather than harvested from rubber trees, its properties and availability remain uniform and independent of agricultural or seasonal disruptions. Electrical Insulation: Certain variants (such as butyl rubber) have low gas and moisture permeability and strong electrical insulating properties. Customizability: Chemists can engineer specific formulas (like Styrene-Butadiene Rubber or neoprene) to match exact performance needs for unique industrial uses.

17. Composition: An alloy of copper and zinc (typically 5% to 45% zinc), sometimes with small amounts of lead, aluminium, or manganese. Colour and Appearance: Bright, reflective yellow-gold (similar to gold). Hardness and Strength: Softer and more malleable (easier to bend, shape, or machine). Conductivity: Higher electrical and thermal conductivity than bronze. Corrosion Resistance: Good overall, though it can struggle or corrode faster in saltwater. It is also naturally antibacterial. Melting Point: Lower, around 900°C – 940°C. Common Uses: Musical instruments (horns, trumpets), plumbing fixtures, electrical connectors, door handles, and decorative items.

18. Fibreboard is an engineered wood product made by breaking down wood chips, shavings, or residual fibres into wood pulp, mixing them with a synthetic resin or glue, and pressing them under high heat and pressure. Low-Density Fibreboard (LDF) / Particleboard / Insulation Board, Medium-Density Fiberboard (MDF), High-Density Fiberboard (HDF) / HardboardHDHMR (High-Density High-Moisture Resistant) Boards. Physical Properties, Mechanical PropertiesUniform Structure: Unlike natural wood, fiberboard lacks knots, grain patterns, or criss-cross layers, giving it a flat,

smooth, and consistent surface. Free of natural grain, fiberboard cuts, drills, and routs cleanly without tearing out or splintering at the edges.

19. Polymerization is a chemical process where small, individual molecules called monomers bond together to form large, high-molecular-weight chain or network molecules called polymers. Addition Polymerization (Chain-Growth): Unsaturated monomers (often containing double bonds, like ethene) link together rapidly without losing any extra atoms or producing byproducts. Condensation Polymerization (Step-Growth): Two different functional monomers combine while releasing small byproduct molecules—such as water (H_2O) or hydrogen chloride (HCl)—at each step. Natural Polymers: Cellulose, starch, natural rubber, and proteins. Synthetic Polymers: Polyvinyl chloride (PVC), polystyrene, nylon, and polyester.

20. Finishing treatments for aluminium, brass, and copper depend on whether your goal is corrosion protection, mechanical smoothing, or decorative coloring. Because these three metals behave differently, they require specific finishing methods. Aluminium is lightweight and forms a natural oxide layer, but it benefits greatly from engineered surface treatments for durabilityAn electrochemical process that thickens the natural oxide layer, improving wear resistance and allowing the metal to be dyed in various colorsProvides a tough, colored polymer layer that seals the metal against harsh environmentsUses abrasives or bead blasting to create a matte, satin, or directional grain finish. A thin chemical film used mostly for corrosion resistance and as a primer for pain. Brass (an alloy of copper and zinc) tarnishes easily in open air, so finishes focus on either preserving its bright golden shine or giving it an aged/blackened look. Mechanical smoothing using fine compounds and buffing pads to achieve a mirror shine, followed by a clear lacquer or wax to prevent oxidation. Immersion in a potassium sulfide or specialized blackening solution to yield an antique, dark bronze, or black patina. Acid dipping to remove scale and free iron from the surface, restoring the bright base alloy. Copper is soft, highly conductive, and develops a natural green-blue patina (verdigris) or brown tarnish over timeUsing chemical agents (like liver of sulfur or specific acid solutions) to artificially age copper into antique brown, black, or teal-green tones. Using chemical agents (like liver of sulfur or specific acid solutions) to artificially age copper into antique brown, black, or teal-green tones. Plating copper with nickel, chrome, or silver to improve surface hardness and change its decorative appearance.

PART -C

21. Steel is a strong, hard metal made by combining iron with carbon (typically between 0.02% and 2.1% by mass) and often other elements like chromium or nickel. Withstands heavy structural loads and resists tension, bending, and deformation better than pure iron. Absorbs high impact energy without fracturing while remaining malleable enough to be shaped, bent, or forged. Increases with higher carbon content or through heat treatment, though higher hardness reduces overall ductility. Primarily iron (97%–99%) combined with carbon and trace elements like silicon, sulphur, phosphorus, and manganese. Primarily iron (97%–99%) combined with carbon and trace elements like silicon, sulphur, phosphorus, and manganese. Readily joined using standard welding methods, governed by its carbon equivalent value. Can be cut, drilled, and shaped depending on the specific grade. Can be melted down and infinitely recycled without losing its core structural value.

    Steel frames, thin stringers, and cable or glass-and-steel balustrades create floating or minimalist staircases. Loft and industrial-style interiors celebrate structural steel frames, I-beams, and pipes as visual "bones" of the space. Laser-cut steel screens, mesh curtains, and frame partitions divide open-plan spaces without blocking light. Slim steel-framed glass doors and internal factory-style windows provide strong, narrow sightlines.

    Dining tables, coffee tables, bed frames, and ergonomic chairs use steel tubing, hairpin legs, or heavy structural plate bases. Industrial open shelving units, wall-mounted

steel brackets, and heavy-duty cabinets offer high weight capacity. Steel pendant lights, minimalist floor lamps, and metallic wall sconces introduce directional or reflected illumination.

22. Selecting the right interior finishes requires balancing aesthetic intent, performance durability, maintenance requirements, and budget constraints. The right choice ensures longevity based on how the space is used. Upholstery must withstand constant friction, weight, and potential spills. General decorative fabrics (like drapery) focus more on drape and light fastness. The Durability Metric, Weave and Density, Material Types. Carpets handle foot traffic, impact insulation, and dictate the acoustic comfort of a room. ideal for high traffic and resistant to abrasion. The premier natural choice. It offers superior pile retention, natural flame retardancy, and a soft texture. Offers excellent stain resistance and vibrant colour fastness, though it can mat down faster than nylon under heavy traffic. Loops are cut straight. It yields a soft, formal look but shows vacuum lines and footprints easily. Uncut loops create a dense surface. Highly durable for hallways and stairs, but can loop-snag if you have pets with claws. Highly twisted fibres that lay in different directions. Great at hiding dirt, tracks, and traffic wear. Wall coverings define the visual plane of a room and hide minor substrate flaws. High-Humidity (Bathrooms/Kitchens): Avoid paper-faced options. Choose Solid Vinyl or vinyl-coated coverings which are scrubbable and resist ambient moisture. Low-Traffic/Formal (Dining rooms/Master bedrooms): Flock, Silk, or Grass cloth textures add depth and organic luxury but cannot handle moisture or heavy scrubbing.
23. Interior furnishing materials are the raw, natural, synthetic, or composite substances used to build, clad, and decorate indoor spaces. They define a room's physical structure, comfort, durability, and style. Hard Furnishing Materials (Structural & Surface Elements), Soft Furnishing Materials (Textiles & Upholstery). Classic hardwoods (like oak, mahogany, and cherry) and softwoods used for furniture, flooring, and architectural trim. Plywood, Medium-Density Fibreboard (MDF), particleboard, and wood veneers that provide strength and smooth surfaces for cabinets and wardrobes without warping like solid wood. Granite, marble, slate, and limestone used for luxury countertops, floors, and feature walls. Steel, iron, aluminium, and copper used for structural framing, industrial-style furniture, railings, and hardware. Float, toughened, or laminated glass used for partitions, tabletops, and modern doors to add openness and transparency. Man-made, durable surface sheets that mimic wood or stone patterns at a lower cost.

    Breathable and versatile, commonly used for bedding, slipcovers, and casual drapes. Lightweight with a natural texture, ideal for curtains and accent pillows. Silk adds a luxurious sheen for low-sun accents, while wool provides warmth and sound insulation in rugs and carpets. Durable, wrinkle-resistant, and stain-resistant fibres often blended into high-traffic upholstery and drapes. Plush or synthetic alternatives used for statement sofas, headboards, and poufs.